MUHAMMED SHAMIL V

Full Stack Developer

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SUMMARY

Full stack developer with experience building end-to-end web applications using the MERN stack and TypeScript. Strong backend foundation with hands-on experience in distributed systems, event-driven architecture, and API design. Comfortable owning features across the entire stack, from React frontend to Node.js backend to database design and deployment.

SKILLS

Frontend: React.js, Redux Toolkit, Tailwind CSS, HTML5, CSS3, Socket.IO (client)
Backend: Node.js, Express.js, REST APIs, JWT, Kafka, Socket.IO, Pino
Databases: MongoDB, PostgreSQL, Redis
Languages: TypeScript, JavaScript
Tools & Infra: Docker, pnpm Workspaces, Drizzle ORM, Git, GitHub

EXPERIENCE

NEXTGENPRO INNOVATIONS AND EDUPARK PVT LTD | Software Engineer | *Feb 2025 – Present*

- Developed full-stack web applications using Node.js, Express, MongoDB, and React for client requirements.
- Designed and built REST APIs for authentication, data handling, and core application features.
- Implemented JWT-based authentication and role-based access control for secure user management.
- Structured backend using modular architecture (controllers, services, middleware) to improve maintainability.
- Designed MongoDB schemas and handled data flow between backend and frontend.
- Debugged and resolved backend issues, improving API stability during development and testing.

PROJECTS

Cab Booking Platform — *Node.js, TypeScript, React.js, Kafka, Redis, PostgreSQL, Docker*

github.com/SHAMIL-VAKKAYIL/cab-booking

- Designed and built a microservices monorepo with 10 services and 21+ Kafka topics covering the full booking lifecycle from ride request to payment and notification.
- Implemented saga orchestration for the booking flow with step-level compensation, handling failures across matching, pricing, and payment services.
- Built driver matching using Redis GeoSet for proximity queries, avoiding database round-trips on the hot path.
- Centralized JWT verification at the API gateway with trusted header forwarding, keeping auth logic out of individual services.
- Designed pricing-service as a stateless Kafka-only service with no HTTP endpoints, using Haversine formula for fare calculation.
- Built shared packages for Kafka messaging with retry and DLQ, typed event contracts, and structured logging across all services.

DevChat — *Node.js, Express.js, Socket.IO, React.js, MongoDB, Redux Toolkit*
github.com/SHAMIL-VAKKAYIL/devChatServer | github.com/SHAMIL-VAKKAYIL/devChatClient

- Built a real-time one-on-one chat application with JWT authentication, image sharing, and persistent message history over WebSocket connections.
- Integrated Socket.IO on both client and server, managing connection lifecycle, room handling, and real-time event synchronization.

EDUCATION

Grace Valley Arts and Science College, Kerala | **Bachelor of Computer Application** | *2021 – 2024*

TRAINING

Techwingsys, Ernakulam | **MERN Stack Development** | *Sep 2024 – Jan 2025*